

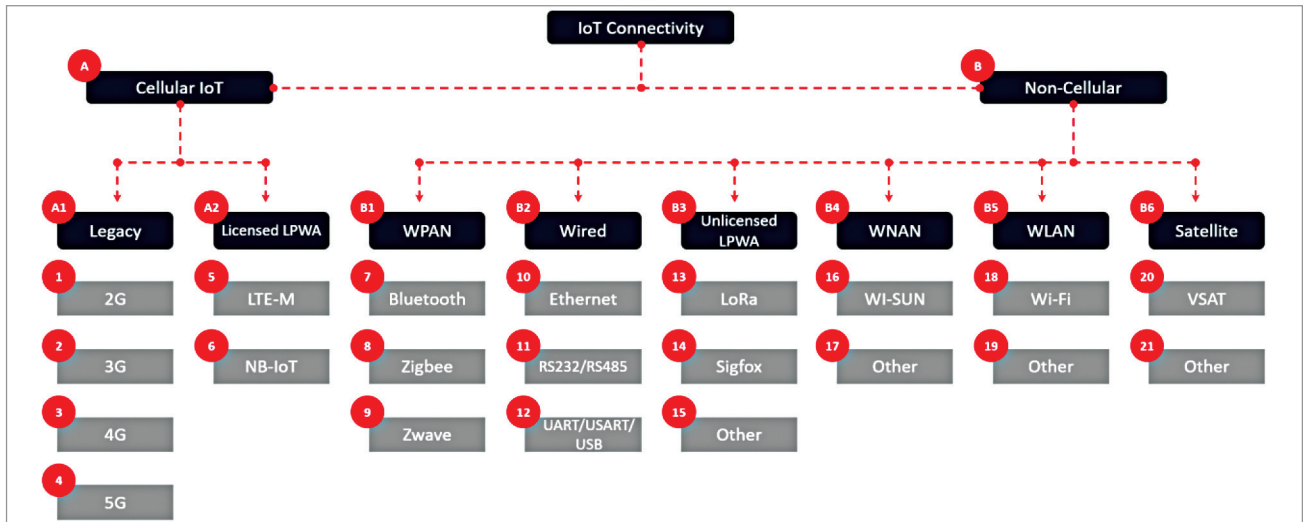


Fig. 3: 21 important IoT connectivity standards

Analytics expects the penetration of IoT MCUs in the general MCU market to grow from 18% in 2019 to 29% by 2025. Particularly 32-bit MCUs, which can be found in devices such as the well-known Raspberry Pi, represent a sweet spot for IoT applications.

2. IoT connectivity chipsets. IoT connectivity chipsets are at the heart of all IoT-connected devices and

represent the largest IoT semiconductor market segment (35% of all IoT semiconductors in 2020). In recent years, the lack of globally accepted IoT connectivity standards and the varying requirements from different use cases and applications have led to a wide variety of connectivity options. IoT Analytics identifies 21 main IoT connectivity standards (from a semi-

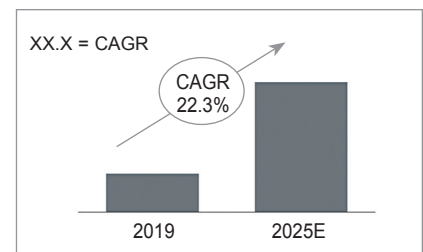


Fig. 4: Global AI IoT chipset growth

Any opportunities you foresee for India's nascent electronics/IoT industry?

In terms of IoT, India is still at nascent stage. However, Jio's launch of NB-IoT nationwide network and partnering with both device players and cloud players makes Jio a potential IoT leader in coming years. In terms of hardware opportunities, India will continue to be dependent on China and other foreign players. However, as we have seen, the Make in India initiative has led to local manufacture and assembly of smartphones. We can expect the same trend for IoT devices in coming years.

What is your take on NB-IoT? Many in the telecom sector have been talking about it, but we are yet to see major roll outs. Is that going to change soon?

NB-IoT is already making headlines. By the end of 2020, globally, 128 operators had deployed or launched NB-IoT or LTE-M networks in 62 countries. Further, 158 operators are actively investing in NB-IoT technology, of which 111 have deployed or commercially launched NB-IoT networks. NB-IoT represented 9% of total IoT connections by 2020, where China contributed 90% in terms of IoT connections. Outside of China, NB-IoT's adoption is slow as LTE-M, LoRa, and now 4G LTE-Cat 1 are constantly competing with NB-IoT. As mentioned, Jio's NB-IoT nationwide network holds much potential to grow India's contribution to IoT.

How's deployment of 5G going to affect the 21+ standards? Are we going to see more, or are we looking at amalgamation/consolidation of some?

Release 17 of 3GPP will consolidate almost all cellular IoT networks into one umbrella of 5G. However, WI-FI, Bluetooth, wired, VSAT, and unlicensed LPWA will remain standalone.

If you were to advise governments wanting to define a policy related to IoT that would enable their start-ups, MSMEs, and MNCs to benefit from IoT and semiconductor play, what would be your advice to them?

I will advise the following:

- Funds to help a start-up to enter the IoT ecosystem
- Fewer duties on import of raw material for semiconductor components
- Tax benefit on local manufacturing
- More government-oriented projects (learning from the Chinese government)

Do you foresee any role of start-ups in this eco-system, or is that a game for 'big money' only?

In IoT, there are many components required to deploy a solution, of which device, network, platform, and cloud are the four major heads. A start-up with a unique solution or technology can create space for itself. However, partnership and collaboration among all verticals is the key to success. Even the big players are not doing everything on their own. However, we can see a lot of takeover of good IoT start-ups by large IoT players.